1 Project Discovery & Planning

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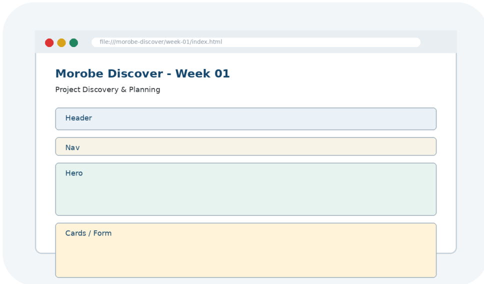

Objectives

You will have mastered the material in this chapter when you can:

- Define the project problem, identify users/tasks, and structure the site architecture.
- Identify how this week builds from the previous project checkpoint.
- Read and interpret key code snippets related to the feature.
- Explain how this week prepares the next project stage.
- Explain the key concepts and terminology for this week.
- Apply the new technique to the Morobe Discover project.
- Test the weekly project increment and record evidence.

Introduction

Project Discovery & Planning introduces the next major layer of the Morobe Discover website. This chapter is written for students who are building the same project from start to final release. The topic is not treated as a separate exercise; it is a practical step in developing one working website.

The site needs direction before coding so every later feature supports a real user task. Professional web development requires students to understand why a technique exists, what problem it solves, and how it changes the behaviour or quality of a webpage.

Project - Morobe Discover

Morobe Discover is a tourism and local discovery website. Each week adds a working layer to the project. This week the project moves from **A website idea only; no project evidence or code structure exists.** to **A documented project plan and repository scaffold ready for semantic HTML.**

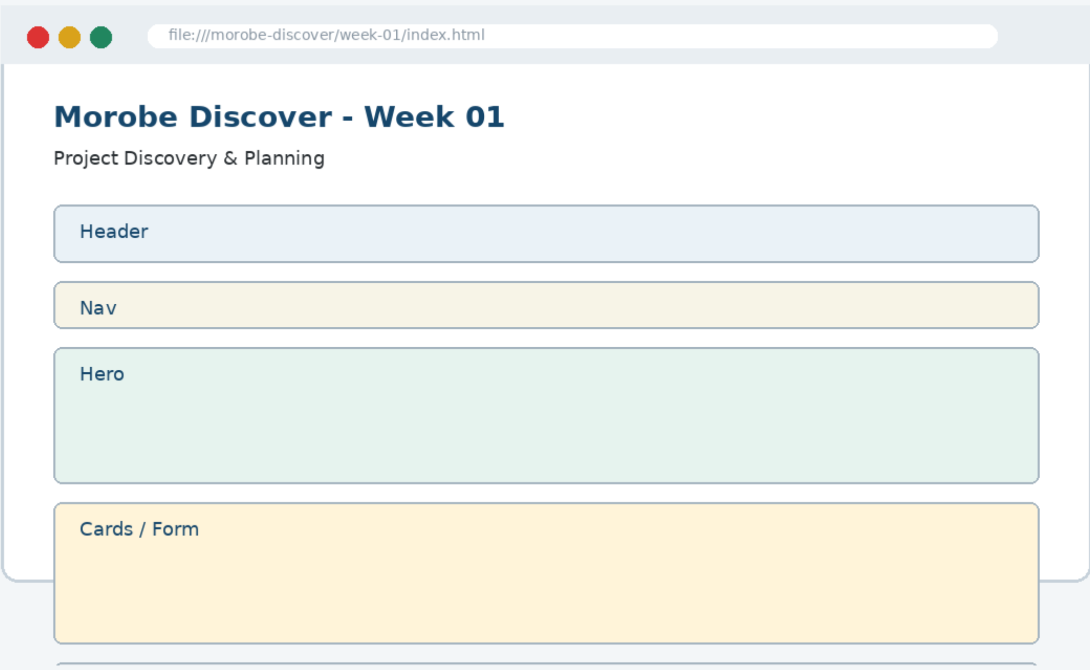

Figure 1-1. Current project focus for Week 01: Project Discovery & Planning.

By the end of the chapter, the project evidence should clearly show the development focus for this week: **Project brief, user stories, sitemap, wireframes, repository scaffold, purpose statement and user personas**.

Before This Week and After This Week

Before Week 01	After Week 01
A website idea only; no project evidence or code structure exists.	A documented project plan and repository scaffold ready for semantic HTML.
Project evidence from the previous checkpoint must be preserved.	The new feature becomes the starting point for Week 02.

Weekly Development Roadmap

1 Review current project	2 Understand the main concept	3 Inspect the interface problem	4 Study the key code
5 Apply it to the project	6 Test the result	7 Commit evidence	8 Prepare for next week

This roadmap mirrors a professional workflow. Students first inspect the current project state, then learn the concept, apply key code, test the visible result and commit the evidence.

1. Core Principle

Plan the problem, the users and the content before choosing colours or writing HTML.

> **Developer Tip:** A good web developer can explain not only what code was written, but why that code is appropriate for the user task and project context.

Concepts Introduced This Week

Concept	Purpose	Project use
Problem statements guide feature decisions	Morobe Discover application	Used in Week 01 project evidence
Personas represent realistic user tasks	Morobe Discover application	Used in Week 01 project evidence
Sitemaps plan pages; wireframes plan page layout	Morobe Discover application	Used in Week 01 project evidence

The concepts in this chapter are introduced in small pieces. Each piece is immediately connected to the Morobe Discover project so that students can see how the idea becomes working project evidence.

2. Explain - Illustrate - Apply

The first step is to understand the interface or coding problem. A weak implementation usually focuses on surface appearance only. A stronger implementation identifies the structural or interaction principle and applies it consistently.

Weak approach

Add code until the page appears to work. Ignore structure, reusability or testing.

Professional approach

Use the correct concept, name the affected component, test the output and commit evidence.

Thinking Question

> **Question:** How does this week improve the website for the user, not just for the developer?
> **Answer:** It makes the current project easier to use, understand, interact with, test or release.

3. Key Code Pattern A

The following code shows the smallest useful pattern for this week's implementation. It is not a complete project file. The complete project belongs in the GitHub repository under **/week-01/**.

```
# Morobe Discover
Week 01 creates planning evidence.

## Output
- Project brief
- User stories
- Sitemap
```

Read the code line by line. Identify the selector, tag, function or command being used. Then identify the visible effect it should produce in the project.

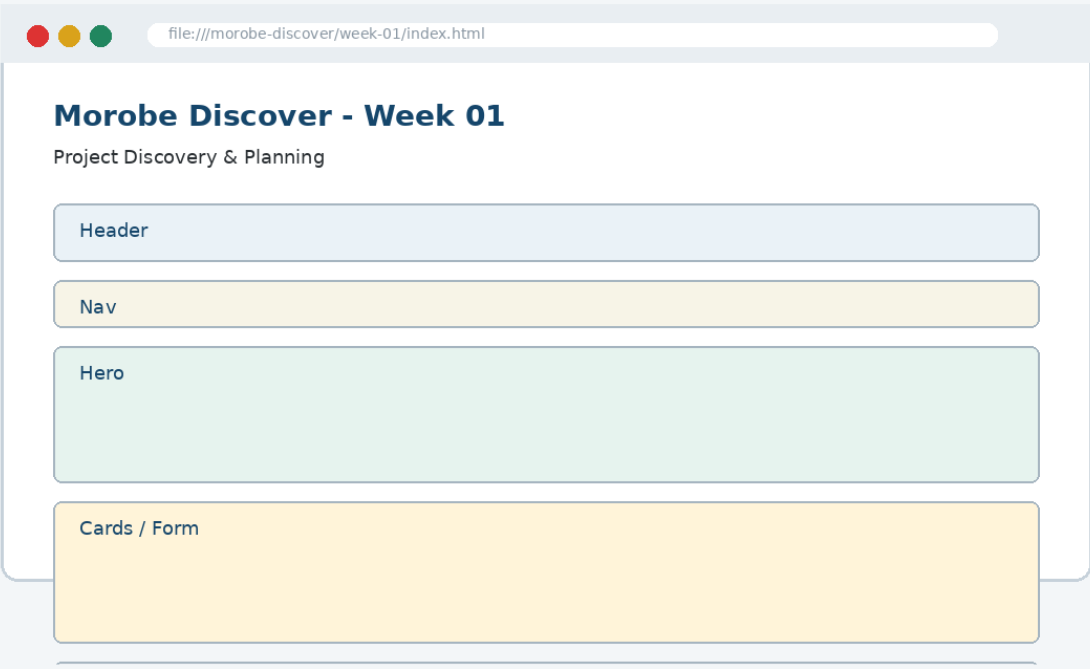

Figure 1-2. Code should be checked against visible output in the browser.

4. Key Code Pattern B

Most weeks require a second pattern that builds on the first. The second pattern usually applies the concept to a different component, state or project file.

```
Home
├── Destinations
├── Events
├── Plan a Trip
└── About
```

Good Practice: Keep code readable, named and testable. Avoid copying code that you cannot explain during a code defence.

What to Inspect

- Which file should contain this code?
- Which part of the browser output should change?
- What error would appear if the code were placed in the wrong file?

5. Applying the Concept to Morobe Discover

Implementation should be completed as a project increment, not as an isolated demonstration. The following steps describe the development path.

To Build the Week 01 Feature

1. Write a purpose statement
2. Create three user personas
3. Write five user stories with acceptance checks
4. Draw the sitemap and homepage wireframe
5. Create README.md and placeholder index.html

Each step should be performed against the same Morobe Discover project folder. Do not create a separate mini-site for this week.

> **Project Rule:** Preserve earlier features. This week adds a layer to the project; it does not replace previous work.

6. Visual Result and Interface Evidence

The browser result is part of the evidence. Students should compare the interface before and after the week to confirm that the new concept has actually changed the site in the intended way.

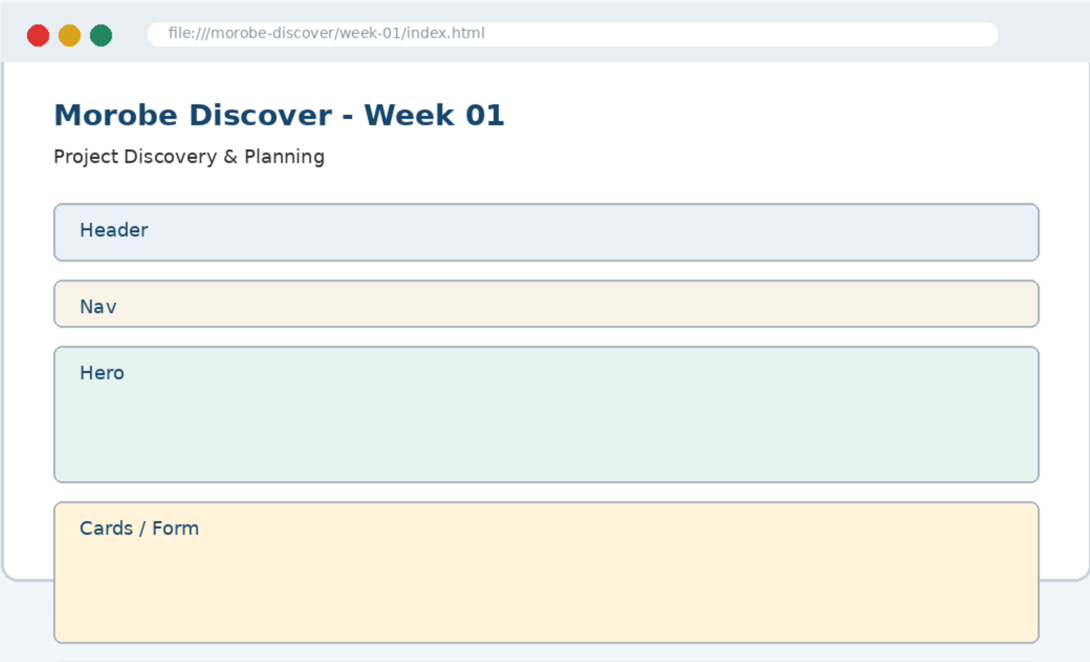

Figure 1-3. Expected Week 01 project output evidence.

Evidence item	What it proves
Screenshot	The feature is visible to the user.
Code file	The implementation exists in the correct location.
Test note	The student checked the behaviour.
Git commit	The project history records the checkpoint.

7. Common Mistake and Correction

Common Mistake: Treating this week's topic as a definition to memorise instead of a technique to apply to the project.

Incorrect approach

The student can define the concept but cannot show where it appears in the Morobe Discover files or browser output.

Correction

Open the relevant project file, locate the code pattern, refresh the browser and explain the output. A student should be able to point to the exact line of code and the exact interface region affected.

```
// Explain during code defence
File: week-01/...
Feature: Project Discovery & Planning
Evidence: browser screenshot + Git commit
```

8. Testing and Validation

Every weekly increment must be tested. Testing confirms that the feature is useful, not just present.

Check	Expected result
Open index.html in the browser	Expected project behaviour is visible and working.
Check user stories include user, goal and reason	Expected project behaviour is visible and working.
Confirm the sitemap matches the planned pages	Expected project behaviour is visible and working.
Commit evidence to Git	Expected project behaviour is visible and working.

Incorrect result and correction: If the page appears unchanged after editing a file, confirm that you edited the correct file, linked the file properly and refreshed the browser. Then inspect the browser console where relevant.

9. GitHub and Project Evidence

The complete implementation for this week should be stored in the project repository, not copied in full into this chapter. The chapter shows key code only so that students focus on understanding.

```
Suggested repository:
https://github.com/YOUR-USERNAME/is229-morobe-discover

Weekly folder:
/week-01/

Complete project:
/complete-project/
```

Commit Message Example

```
git add .
git commit -m "Week 01 Project Discovery & Planning project increment"
```

10. Completed Weekly Outcome

At the end of Week 01, the project should demonstrate the following completed outcome:

> **Completed outcome:** Project brief, user stories, sitemap, wireframes, repository scaffold, purpose statement and user personas

Definition of Done

Done when...	Evidence
The project includes the Week 01 feature.	Updated files in /week-01/
The feature can be explained using key concepts.	Student code explanation
The feature works in the browser.	Screenshot and test note
The work is version-controlled.	Git commit / repository history

11. Bridge to the Next Week

Week 02 converts the planned wireframe regions into HTML5 semantic page structure.

Week 01 output	Next use
Project brief, user stories, sitemap, wireframes, repository scaffold, purpose statement and user personas	Week 02 converts the planned wireframe regions into HTML5 semantic page structure.

This bridge is important because the course uses one cumulative project. The work produced this week becomes part of the starting point for the next class.

Chapter Summary

This chapter introduced Project Discovery & Planning as a practical development step in the Morobe Discover project. You learned the principle behind the topic, studied key code patterns, applied the concept to the project, tested the result and prepared evidence for GitHub.

Apply What You Learned

1. Given “make a tourism website”, rewrite it as a testable project problem.
2. Identify the exact project file that should change this week.
3. Explain how the browser output proves the feature works.
4. Write one test note for the weekly feature.
5. Explain how this week connects to the next week.

> **Before the next class:** Read the next week's material and bring your current project with this week's feature completed and committed.